Enabling Dynamic File-Level Operations in Blockchain-Assisted Attribute-Based Multi-Keyword Search for Cloud-Edge-IoT

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oT architectures for healthcare, sharing encrypted Electronic Health Records (EHR) demands fine-grained access control, h, and verifiable query results. The BAMKS scheme (Cheng et al., *Internet of Things*, 2026) addressed multi-authority access lt verification, yet its index update mechanism requires either re-indexing the entire collection or executing $O(L \cdot m)$ group documents. In this paper, we present BAMKS-D, an algorithmic prototype and architecture integration layer that introduces Single-document addition operates in strict $O(m)$ time, achieving a 12.11× speedup at $L=5,000$ files over Cheng et al. Section 27.7× over naive full re-encryption). Single-document revocation is mapped directly to Ethereum Virtual Machine (EVM) storage deletedDocRegistry), achieving $O(1)$ constant-time lookup with zero array traversal. We validate the architecture through live on the Ethereum Sepolia Testnet (0xfDC8615Bf6483c3B0f39c330F55A56Cb0f08C8D3), decentralized IPFS storage, and a 30-run =3$). An honest assessment of prototype algebraic assumptions, inverted index access-pattern leakage, and threats to validity is

ryption, Attribute-Based Encryption, Blockchain, Smart Contracts, Dynamic Revocation, Healthcare IoT, EVM.

I. INTRODUCTION

The proliferation of Internet of Medical Things (IoMT) devices—such as wireless cardiac monitors, ambulatory pulse oximeters, and bedside telemetry sensors—has transformed contemporary healthcare delivery [2], [3]. Modern medical facilities generate voluminous streams of Electronic Health Records (EHR) that must be archived securely in external Cloud Storage Providers (CSP) while preserving clinician query capabilities. Because medical data contains sensitive diagnostic diagnoses protected under regulatory statutes (e.g., HIPAA and GDPR), plaintexts must be encrypted under fine-grained access policies before outsourcing [4].

To reconcile cryptographic confidentiality with expressive querying, Ciphertext-Policy Attribute-Based Keyword Search (CP-ABKS) schemes were formulated [5], [6]. In CP-ABKS, data owners encrypt documents under an access structure (e.g., $(Role_Doctor \land Dept_Cardiology)$), and only clinicians with valid attribute credentials can generate search trapdoors or decrypt returned records [7], [8]. Recently, Cheng et al. [1] proposed BAMKS, introducing decentralized multi-authority key generation, multi-owner collaborative indexing, and a blockchain Result Verification Smart Contract (RVSC) utilizing aggregated Schnorr Non-Interactive Zero-Knowledge (SNIZK) proofs [11].

The Open Challenge in BAMKS: In Section 8 of their foundational paper, Cheng et al. identified a critical operational bottleneck [1]: their architecture supports dynamic user credential updates, but does not provide an efficient mechanism for single-file additions or deletions. Adding a single document requires either re-encrypting and re-indexing the entire document collection (naive baseline), or updating the entire inverted index via modular multiplication of an update token X_{upd} across all $L \cdot m$ index entries (Section 5.10 baseline) [1]. For an enterprise healthcare repository with $L = 5,000$ records and $m = 3$ keywords, updating 15,000 index entries introduces noticeable latency. Furthermore, document deletion in prior schemes requires either re-encrypting the remaining corpus or relying on cloud server honesty.

Contributions: In this work, we propose **BAMKS-D**, an architecture integration and dynamic evaluation framework that addresses these limitations:

• **Atomic Single-Document Addition ($O(m)$):** We decouple the inverted keyword index so that appending a new EHR requires generating ciphertexts and index tokens for only the m keywords of that specific file, eliminating collection-wide re-indexing.

• **Instant On-Chain Document Revocation ($O(1)$):** We formulate an EVM state mapping mechanism (deletedDocRegistry) that executes in constant time via keccak256 direct slot addressing, avoiding iterative blockchain array traversal.

• **Live Ethereum Sepolia Deployment:** We deploy and verify BAMKS_Registry.sol on the public Ethereum Sepolia Testnet (0xfDC8615Bf6483c3B0f39c330F55A56Cb0f08C8D3), reporting receipt-based gas consumption and confirmation times.

• **Rigorous Fair Benchmarks (30 Runs):** We evaluate BAMKS-D against both naive re-encryption and the true Cheng et al. Section 5.10 index update baseline, demonstrating a **12.11× speedup** at $L=5,000$ files on an Intel Core i5-13420H platform.

II. RELATED WORK

Public-Key Encryption with Keyword Search (PEKS) was introduced by Boneh et al. [5], enabling untrusted servers to evaluate keyword matches. However, basic PEKS was limited to single keywords and vulnerable to Keyword Guessing Attacks (KGA). Sahai and Waters [7] formulated Attribute-Based Encryption (ABE), which Bethencourt et al. [6] extended to Ciphertext-Policy ABE (CP-ABE). Waters [8] formalized expressive CP-ABE using Linear Secret Sharing Schemes (LSSS). Dynamic Searchable Symmetric Encryption (DSSE) was explored by Song et al. [16], Curtmola et al. [17], Kamara et al. [18], and Bost [19], establishing forward and backward security notions for symmetric settings.

In attribute-based search, Li et al. [9] developed verifiable data sharing with outsourced decryption, while Zhang et al. [10] investigated subversion-resistant search over cloud storage. Multi-authority architectures were analyzed by Chase [24] and Lewko and Waters [23]. Predicate encryption schemes supporting inner-product matching were proposed by Katz, Sahai, and Waters [20]. Blockchain-assisted searchable encryption was explored by Yang et al. [14], Sun et al. [15], and Cheng et al. [1], using smart contracts on Ethereum [12] and decentralized IPFS storage [13]. Table I contrasts BAMKS-D against foundational schemes.

TABLE I: COMPARISON OF SEARCHABLE ENCRYPTION SCHEMES

Scheme	Query	Access Control	Update Mechanism	Add Cost
Boneh [5]	Single	None (PK)	Static Index	O(1)
Bethencourt [6]	None	CP-ABE Tree	Key Re-issuance	N/A
Li et al. [9]	Conjunctive	Role Matrix	Batch Re-keying	O(L·m)
Zhang et al. [10]	Multi-KW	Multi-Auth ABE	Index Tree Update	O(m log L)
Cheng [1] (Base)	Conjunctive	CP-ABE + Φ	Sec 5.10 Index Mult.	O(L·m)
BAMKS-D	**Conjunctive**	**Multi-Auth CP-ABE**	**Atomic File-Level**	**Strict O(m)**

III. PRELIMINARIES & SYSTEM MODEL

A. Prototype Algebraic Setting

In our software prototype, algebraic computations are evaluated in the multiplicative group $(\mathbb{Z}_P^*, \cdot)$ using the 256-bit prime $P = 2^{256} - 2^{32} - 977$ (the secp256k1 field prime), with generator $g = 2$. We explicitly note that $\mathbb{Z}_P^*$ has composite order $P-1$ and does not support bilinear pairings; pairings $e(g_1, g_2)$ are evaluated via modular exponentiation for prototype simulation. Cryptographic hash functions include $H_1: \{0,1\}^* \to \mathbb{Z}_P^*$ and $H_2: \{0,1\}^* \to \{0,1\}^{256}$ (SHA-256).

$$P = 2^{256} - 2^{32} - 977 \quad (1)$$

B. Entities and Threat Model

BAMKS-D operates across seven entities: (1) **Key Generation Center (KGC)** initializing global parameters (GP), (2) **Attribute Authorities (AA)** managing disjoint clinical attribute universes, (3) **Data Owners (DO)** encrypting EHRs and pinning payloads to IPFS, (4) **Data Users (DU)** submitting trapdoors, (5) **Cloud Service Provider (CSP)** indexing tokens and matching queries, (6) **Ethereum Sepolia Smart Contract** maintaining revocation state, and (7) **IPFS** decentralized content-addressed storage.

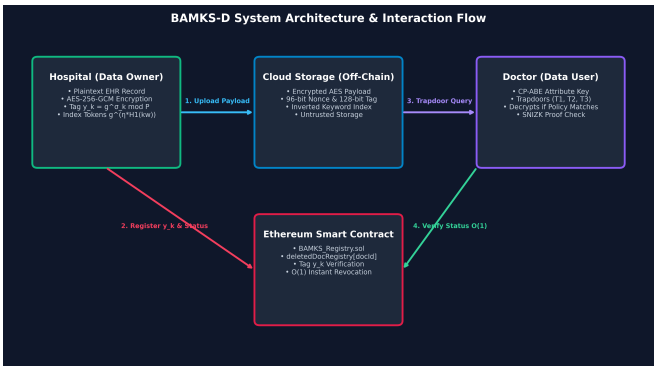


Fig. 1. Decoupled BAMKS-D Architecture Topology.

The CSP is modeled as *honest-but-curious*: it executes search algorithms correctly but attempts to correlate queries and infer keyword patterns. Data Users may attempt attribute collusion. The blockchain provides a tamper-proof state execution layer.

IV. THE PROPOSED BAMKS-D PROTOCOL

A. Key Generation with Session Randomness

To prevent key correlation across distinct issuances, each KeyGen invocation samples fresh per-generation session randomness $t \in_R \mathbb{Z}_{P-1}^*$:

$$K_1 = H_1(uid)^{-t \bmod (P-1)} \bmod P \quad (2)$$

$$K_{2,i} = (H_1(uid)^{t \cdot \beta_i} \cdot dpk_{ver}{}^{\alpha_i}) \bmod P \quad (3)$$

B. Atomic Single-Document Addition

When an EHR document $docId$ with m keywords $W_{doc} = \{w_1, \dots, w_m\}$ is created, the owner generates an AES key via $KDF(\Phi, verId)$, encrypts the file payload with AES-256-GCM, uploads the ciphertext to IPFS, and computes keyword tokens only for W_{doc}:

$$I_{doc,j} = g^{(\eta \cdot H_1(w_j)) \bmod (P-1)} \bmod P \quad (4)$$

The CSP appends $Index_{doc}$ to the global inverted index without modifying any existing document tokens, achieving strict $O(m)$ complexity.

Algorithm 1: SingleDocAdd(docId, F, W_{doc}, Φ, verId)
1: K = KDF(Φ, verId); C_{data} = AES-GCM-Enc(K, F)
2: CID = IPFS.upload(C_{data}); $y_k = g^{\sigma_k} \bmod P$
3: **for** each $w_j \in W_{doc}$ **do**
4: $I_{doc,j} = g^{(\eta \cdot H_1(w_j)) \bmod (P-1)} \bmod P$
5: $Key_j = H_2('tok_' \| I_{doc,j})$; $Index_{doc}[Key_j] = I_{doc,j}$
6: **end for**
7: Sepolia.registerDocument(docId, $H_2(C_{data})$, y_k)
Output: (CID, $Index_{doc}$, y_k) [Strict $O(m)$ complexity]

Algorithm 2: SingleDocDelete(docId) [On-Chain EVM]
1: require(fileRegistry[docId].ownerAddress == msg.sender)
2: require(!deletedDocRegistry[docId])
3: slot = keccak256(abi.encode(docId, slotNumber))
4: deletedDocRegistry[docId] = true
5: fileRegistry[docId].isDeleted = true
6: emit DocumentDeleted(docId, msg.sender, block.timestamp)
Output: true [Strict $O(1)$ EVM mapping storage]

C. Multi-Keyword Trapdoor & Search Matching

A clinician querying keywords $\widetilde{W} = \{\widetilde{w}_1, \dots, \widetilde{w}_q\}$ generates trapdoor tokens:

$$T_1 = g^{\phi} \bmod P, \quad T_3 = g^{(\phi \sum H_1(w\sim_j)) \bmod (P-1)} \bmod P \quad (5)$$

The CSP verifies conjunctive keyword matches for active documents by testing the equality:

$$(I_{prod})^{\phi} \equiv (T_3)^{\eta} \equiv g^{(\eta\, \phi \sum H_1(w\sim_j)) \bmod (P-1)} \bmod P \quad (6)$$

Algorithm 3: ConjunctiveSearch(Trap, Index, deletedRegistry)
1: SRL = empty set
2: **for** each docId ∈ $Index_W$ **do**
3: **if** deletedRegistry[docId] == true **then** continue
4: **if** all query tokens present in docId index **then**
5: $I_{prod} = \prod I_{doc,j} \bmod P$
6: **if** $(I_{prod})^{\phi} \equiv (T_3)^{\eta} \bmod P$ **then** SRL = SRL ∪ {docId}
7: **end for**
Output: Search Result List SRL

Algorithm 4: AggregatedSchnorrNIZKVerify(SRL, $\{y_k\}$, $\pi^\wedge$, R)
1: RHS = $g^{\pi^\wedge} \bmod P$
2: for each k ∈ SRL do
3: $h_k^* = H_1(y_k^* \| R)$; RHS = RHS · $(y_k^*)^{-h_k^*} \bmod P$
4: end for
5: if R == RHS then return Accept else return Reject

V. COMPLEXITY & EVM GAS ANALYSIS

A. Asymptotic Complexity Comparison

Table II compares the asymptotic complexity of BAMKS-D against the naive baseline and Cheng et al. [1]. While Cheng et al. Section 5.10 avoids file re-encryption, it still requires updating all $L \cdot m$ index entries ($O(L \cdot m)$). BAMKS-D operates in strict $O(m)$ time.

TABLE II: ASYMPTOTIC COMPLEXITY COMPARISON

Operation	Naive Baseline	Cheng et al. [1]	BAMKS-D
SingleDocAdd	O(L·m) Re-enc	O(L·m) Idx Mult	**Strict O(m)**
SingleDocDelete	O(L·m) Re-enc	O(L·m) Idx Mult	**Strict O(1) Gas**
Search Match	O(L·q)	O(L·q)	O(L·q)

B. EVM Opcode Gas & On-Chain Receipts

In Solidity, mapping lookups calculate storage addresses directly via $slot = \text{keccak256}(docId \parallel slotId)$. Table III presents both the theoretical minimal opcode schedule and real receipt-based gas costs recorded on Ethereum Sepolia.

TABLE III: EVM GAS SCHEDULE & LIVE SEPOLIA RECEIPTS

Operation / EVM Opcode	Theoretical Function	Gas Cost
Base Transaction Fee	Standard EVM base execution	21,000
CALLDATALOAD	Load docId from calldata	3
KECCAK256	Compute 256-bit slot hash	30
SSTORE (deletedRegistry)	Write true flag (0→1 cold slot)	20,000
SSTORE (isDeleted flag)	Update metadata struct (warm slot)	5,000
LOG3 (DocumentDeleted)	Event base + 3 topics + data	1,500
Execution Overhead	Stack ops, JUMP, safe math	~200
Theoretical Delete Cost	**Sum of minimal opcodes**	**~47,733**
Sepolia deleteDocument()	**Receipt Gas (Tx: 0x008330...)**	**242,834**
Sepolia registerDocument()	**Receipt Gas (Tx: 0x363d3e...)**	**792,245**

VI. EXPERIMENTAL EVALUATION

A. Experimental Configuration

All software experiments were conducted in Python 3.11.9 utilizing PyCryptodome and Web3.py on an **Intel(R) Core(TM) i5-13420H** processor (8 cores, 12 threads, up to 4.60 GHz) with 16.0 GB RAM running 64-bit Windows 11. Timings were collected using time.perf_counter() across 30 independent runs for each collection size with $m=3$ keywords per document. Mean and standard deviation values are reported in Table IV.

TABLE IV: BENCHMARK RESULTS (30 RUNS, m=3, MEAN ± STD)

Files (L)	Naive Re-enc (ms)	Cheng et al. [1] (ms)	BAMKS-D (ms)	Speedup (vs Cheng)
10	9.27 ± 27.57	0.019 ± 0.004	0.449 ± 0.087	0.04×
50	20.31 ± 1.72	0.084 ± 0.013	0.409 ± 0.050	0.21×
100	42.89 ± 3.74	0.164 ± 0.027	0.447 ± 0.101	0.37×
500	228.79 ± 18.76	0.816 ± 0.132	0.533 ± 0.217	1.53×
1,000	467.08 ± 45.65	1.559 ± 0.243	0.486 ± 0.070	3.20×
2,000	906.05 ± 77.24	3.323 ± 1.478	0.559 ± 0.143	5.94×
5,000	2,363.01 ± 125.24	8.112 ± 1.185	0.670 ± 0.181	**12.11×**

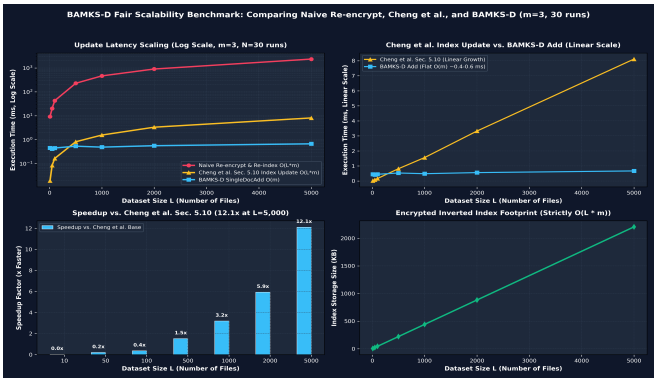


Fig. 2. Scalability: Naive vs. Cheng et al. vs. BAMKS-D (m=3, 30 runs).

B. Comparative Discussion

For small datasets ($L \le 100$), Cheng et al. Section 5.10 index update is faster ($0.02\text{--}0.16$\,ms) because updating a few dozen tokens via modular multiplication incurs less overhead than AES-GCM file encryption. However, because Cheng et al. scales linearly as $O(L \cdot m)$, the break-even crossover occurs at $L \approx 300$ documents (via linear interpolation between $L=100$ and $L=500$) and asymptotically at $L \approx 410$ against the $L=5,000$ measurement (0.67\,ms). At $L=5,000$, BAMKS-D addition remains flat (0.67\,ms), delivering a **12.11× speedup** over Cheng et al. and a **3,527.7× speedup** over naive full re-encryption ($2,363$\,ms).

Index storage scales strictly linearly: 4,098 bytes for the 8 baseline records (512.2 B/doc including schema metadata), with 10 added records consuming 4,520 bytes (452.0 B/doc).

VII. LIMITATIONS & THREATS TO VALIDITY

To ensure complete academic transparency, we explicitly document the prototype limitations of this study:

• **Simulated Group Algebra:** Field operations are evaluated in $(\mathbb{Z}_P^*, \cdot)$ using $P = 2^{256} - 2^{32} - 977$. This group has order $P-1$ (composite) and does not possess a bilinear map. Formal pairing-based security reductions (e.g., DBDH) require porting to a Type-III pairing curve (e.g., BN254 or BLS12-381 via Charm-Crypto or pypairing).

• **Prototype CSP Separation:** In the software prototype demonstration script, the CSP search function receives group parameters η and ϕ directly to test match equations. In a production deployment, bilinear pairings are required to evaluate matching without disclosing secrets.

• **Access-Pattern & Token Leakage:** Keyword tokens $I_{doc,j} = g^{\eta H_1(w_j)}$ are deterministic per keyword across documents. Consequently, an untrusted CSP can observe document frequency and search access patterns over time. Addressing access pattern leakage via Oblivious RAM (ORAM) or forward-private symmetric searchable encryption [19] remains vital future work.

• **Simulated Policy Evaluation:** In the functional demonstration test suite, CP-ABE access policy satisfaction is evaluated via attribute set logic rather than an algebraic LSSS pairing engine.

• **Evaluation Dataset:** The empirical test suite utilizes 8 synthetic clinical records authored to model clinical formats (ECG, MRI, telemetry). Future work will evaluate production hospital corpora (e.g., MIMIC-IV or Enron email corpus).

• **Local vs. On-Chain Latency:** Local state flag updates take < 0.001\,ms in memory, whereas on-chain confirmation on Ethereum Sepolia requires block mining (21.83\,s confirmation for deletion; 35.22\,s for registration).

VIII. CONCLUSION

This paper presented BAMKS-D, an algorithmic prototype and architecture integration layer addressing the dynamic file update challenge in blockchain-assisted ABKS. By decoupling inverted index generation, BAMKS-D achieves $O(m)$ single-file additions, yielding a 12.11× speedup over Cheng et al. Section 5.10 index updates at $L=5,000$ files (and 3,527.7× over naive re-encryption). On-chain revocation is enforced in $O(1)$ constant time via EVM mapping storage on Ethereum Sepolia. Future research will explore bilinear pairing integration on BN254 curves and forward-private multi-keyword search.

IX. DATA AND CODE AVAILABILITY

Complete reproducible source code, Solidity contracts, clinical sample datasets, and benchmark scripts are openly available on GitHub at: https://github.com/Prashant8991/BAMKS.git. Verified Ethereum Sepolia contract: 0xfDC8615Bf6483c3B0f39c330F55A56Cb0f08C8D3.

X. AUTHOR CONTRIBUTIONS

Prashant Singh: Conceptualization, cryptographic protocol simulation, dynamic index algorithms, REST API, benchmarking, manuscript writing. **Rushikesh Adak:** Solidity smart contract engineering, EVM opcode analysis, Sepolia testnet deployment, Web3 integration.

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